

TRINH TRAN (JIMMY)

Researcher in Physiological Health Modeling

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Research scientist with 5 years applying statistical and machine learning methods to physiological and occupational health data. Currently completing an M.S. in Computer Science at Georgia Tech, with research interests in the application of analytics modeling on health outcomes.

EDUCATION

2025 – 2027 (expected) **Georgia Institute of Technology**

M.S. Computer Science (Online), AI Specialization

Relevant coursework: CS 7637 Knowledge-Based Artificial Intelligence (KBAI) — case-based reasoning, knowledge representation, and context-aware AI agents; emphasis on building intelligent systems that reason, learn, and explain decisions.

2015 – 2019 **National University of Singapore**

B.Eng. Biomedical Engineering

Relevant coursework: BN3401 Biostatistics for Biomedical Engineers — hypothesis testing, linear and logistic regression, ANOVA, survival analysis, and study design for clinical and physiological data.

2011 – 2014 **Anglo-Chinese School (Independent)**

International Baccalaureate Diploma

WORK EXPERIENCE

2021 – 2026 **National University of Singapore, YLL School of Medicine**

Research Assistant, Human Potential Translational Research Programme

- Developed and maintained data pipelines across multi-site studies; handled signal cleaning, missing data, and sensor validation to support downstream predictive ML models.
- Built a real-time dashboard for live heat strain monitoring across deployed firefighters; designed for time-sensitive operational decision-making.
- Collected and analysed physiological data across lab and field studies in Singapore, Vietnam, and South Korea, spanning military, firefighter, construction, and delivery worker cohorts.

2020 – 2021 **Kaphos Software Pte Ltd**

Software Developer

- Managed client accounts end-to-end, from requirements gathering to development and delivery.
- Built admin and user-facing web applications; handled both front-end and back-end development.

2018 **Hill-Rom Singapore**

Intern, Global Technology Integration

- Explored upstream medical technology concepts for R&D; contributed to UI/UX bug triage and engineering handoff.

PUBLICATIONS & CONFERENCE PROCEEDINGS

- “External auricle temperature enhances ear-based wearable accuracy during physiological strain monitoring in the heat,” Tan, S.C.C., **Tran, T.C.K.**, Chiang, C.Y.N., Pan, J., & Low, I.C.C. (2024). *Scientific Reports (Nature)*, vol. 14, art. 12689. [doi:10.1038/s41598-024-63241-2](https://doi.org/10.1038/s41598-024-63241-2)
- “Machine learning for heat stress monitoring using environmental data loggers,” **Tran, T.C.K.**, Leow, C.W.H., Tan, P.S.M., Nelstrom, T., & Lee, J.K.W. (2024). *International Conference of Environmental Ergonomics* (poster presentation).
- “Enhancing physical performance with machine learning,” **Tran, T.C.K.**, Wang, Y.R., Remus, A., Low, I.C.C., Ho, D., & Lee, J.K.W. (2022). In *Applications of Machine Learning in Digital Healthcare*, Institution of Engineering and Technology. [doi:10.1049/PBHE036E_ch9](https://doi.org/10.1049/PBHE036E_ch9)

Manuscripts in preparation (placeholder titles)

- “Evidence-based heat stress advisory for vulnerable populations,” Project Cooling Singapore 2.0, (*first author, in collaboration with Ioannou, L., Kingma, B., Gao, C., Morabito, M., et al.*)
- “Heat strain profiling in postal delivery workers under piece-rate employment,” Project HeatSafe Korea, (*co-author*)
- “Heat strain profiling in shoe factory workers”, Project HeatSafe Vietnam, (*co-author*)
- “Heat strain profiling in construction workers”, Project HeatSafe Vietnam, (*co-author*)
- “Wearables for firefighting training and operations”, Project HTX–SCDF firefighter heat strain monitoring, (*co-author*)

SCHOLARSHIPS & AWARDS

2015 – 2019	ASEAN Undergraduate Scholarship , National University of Singapore <i>Full scholarship awarded to outstanding ASEAN undergraduates.</i>
2011 – 2014	A*STAR Scholarship <i>Singapore Government full scholarship for top ASEAN students under the Integrated Programme.</i>
2023	First Prize, NUS CoDesign Competition <i>Designed a plant-growing app to encourage active mobility and socialisation in older adults.</i>
2018	Director’s Award for Biomedical Design Project, NUS <i>Designed an air-compressed implant to delay the onset of early-stage knee osteoarthritis.</i>
2017	Bronze Award (Cash Prize), Design Category, Biomedical Engineering Society <i>Prototyped a wearable biosensor for acute stress monitoring (Arduino).</i>

VOLUNTEERING

2016 – 2023	The Boys’ Brigade, Singapore <i>Volunteer Officer</i> ● Mentored ~40 youths in leadership skills, organisational competencies, and personal values. ● Developed and delivered training curriculum; digitalised internal procedures using automation platforms.
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